Raaz Dwivedi

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Academic appointments

Chief Technology Officer and Cofounder, Traversal *2024—*

Assistant Professor, Operations Research & Information Engineering (ORIE) *2024—*
Field Member: Applied Math, Computer Science, ORIE, Statistics
Cornell Tech, Cornell University

Visiting Assistant Professor, ORIE, Cornell University *Fall 2023*

FODSI Postdoctoral Fellow, CS, Statistics, EECS *2021–2023*
Harvard University & Massachusetts Institute of Technology (MIT)
Advisors: *Prof. Susan Murphy & Prof. Devavrat Shah*

Education

Ph. D., EECS, University of California (UC), Berkeley *2015—2021*
Advisors: *Prof. Martin Wainwright & Prof. Bin Yu*
Thesis title: *Principled statistical approaches for sampling and inference in high dimensions*

B. Tech., EE, Indian Institute of Technology (IIT), Bombay, India *2010—2014*
Minors in mathematics, Institute Rank 1

Research Interests

My research involves a multi-disciplinary approach to data science and brings together ideas from computer science, electrical engineering, and statistics in collaboration with domain experts. I develop statistical machine learning approaches for data-driven personalized decision-making with research across *causal inference, reinforcement learning, Bayesian inference, random sampling, and high-dimensional statistics.*

Selected Achievements & Awards

- Delta Rising Professor, Delta *2026*
- Blackwell Rosenbluth Award for contributions to Bayesian statistics *2024*
- Best Student Paper Award, Statistical Computing & Graphics, American Statistical Association *2022*
- Best Presentation Award, Laboratory of Information & Decision Systems (LIDS) Conference, MIT *2022*
- Certificate of Distinction and Excellence in Teaching (Q Award), Harvard University *2022*
- Foundations of Data Science (FODSI) Postdoctoral Fellowship *2021*
- Outstanding Graduate Student Instructor Award, UC Berkeley *2020*
- Berkeley Fellowship, the most prestigious fellowship for incoming Ph. D. students *2015*
- President of India Gold Medal, IIT Bombay, for the highest GPA in the graduating class *2014*
- All India Rank 10 amongst half a million, IIT Joint Entrance Exam *2010*

Pre-Prints

(title is hyperlinked to the online pdf of the paper)

P1. Albert Gong, Kyuseong Choi, Abhineet Agarwal, Jason Schechner, Ryan Huang, Raj Agrawal, Anish Agarwal, **Raaz Dwivedi**, "ORCA-bench: How ready are language model agents for oncall?", *arxiv.2026*

P2. Albert Gong, Annabelle Michael Carrell, **Raaz Dwivedi**, Lester Mackey, "Express language modeling", *arxiv.* *2026*

P3. Brian M. Cho, **Raaz Dwivedi**, Nathan Kallus, "GAAVI: Global asymptotic anytime-valid inference for the conditional mean function", *arxiv.* *2026*

P4. Kyuseong Choi, Dwaipayan Saha, Woojeong Kim, Anish Agarwal, **Raaz Dwivedi**, "GOPO: Policy optimization using ranked rewards", *arxiv.* *2026*

P5. Tathagata Sadhukhan, Manit Paul, **Raaz Dwivedi**, "Adaptively-weighted nearest neighbors for matrix completion", *arxiv*. *2025*

P6. Jacob Feitelberg, Dwaipayan Saha, Kyuseong Choi, Zaid Ahmad, Anish Agarwal, **Raaz Dwivedi**, "One pipeline, many transformers: Pattern-specific imputation specialists for tabular missing data", *arxiv*. *2025*

P7. Navonil Deb, **Raaz Dwivedi**, Sumanta Basu, "Counterfactual forecasting for panel data", *arxiv*. *2025*

P8. Jacob Feitelberg, Kyuseong Choi, Anish Agarwal, **Raaz Dwivedi**, "Distributional matrix completion via nearest neighbors in the Wasserstein space", *arxiv*. *2024*

P9. Kyuseong Choi, Jacob Feitelberg, Anish Agarwal, **Raaz Dwivedi**, "Learning counterfactual distributions via kernel nearest neighbors", *arxiv*. *2024*

P10. Alberto Abadie, Anish Agarwal, **Raaz Dwivedi**, Abhin Shah, "Doubly Robust Inference in Causal Latent Factor Models", *arxiv*. *2024*

P11. **Raaz Dwivedi**, Katherine Tian, Sabina Tomkins, Predrag Klasnja, Susan Murphy, Devavrat Shah, "Doubly robust nearest neighbors in factor models", *arxiv*. *2023*

P12. Abhin Shah, **Raaz Dwivedi**, Devavrat Shah, Greg Wornell, "On counterfactual inference with unobserved confounding", *arxiv*. *2022*

Conference Publications

C1. Annabelle Michael Carrell, Albert Gong, Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey, "Low rank thinning", *International Conference on Machine Learning (ICML)*. *2025*

C2. Tathagata Sadhukhan, Manit Paul, **Raaz Dwivedi**, "On adaptivity and minimax optimality of two-sided nearest neighbors", *International Conference on Artificial Intelligence and Statistics (AISTATS)*. *2025*

C3. Albert Gong, Kyuseong Choi, **Raaz Dwivedi**, "Supervised Kernel Thinning", *Advances in Neural Information Processing Systems (NeurIPS)*. *2024*

C4. Lingxiao Li, **Raaz Dwivedi**, Lester Mackey, "Debiased Distribution Compression", *International Conference on Machine Learning (ICML)*. *2024*

C5. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey, "Compress then test: Powerful kernel testing in near-linear time", *Conference on Artificial Intelligence and Statistics (AISTATS)*. *2023*

C6. **Raaz Dwivedi**, Lester Mackey. "Generalized kernel thinning", *International Conference on Learning Representations (ICLR)*. *2022*

C7. Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey. "Distribution compression in near-linear time", *International Conference on Learning Representations (ICLR)*, **Best Student Paper Award, JSM**. *2022*

C8. **Raaz Dwivedi**, Lester Mackey, "Kernel thinning", Extended abstract in *Conference on Learning Theory (COLT). Full version under review in JMLR.* *2021*

C9. **Raaz Dwivedi**, Nhat Ho, Koulik Khamaru, Martin J. Wainwright, Michael I. Jordan, Bin Yu, "Sharp analysis of Expectation-Maximization for weakly identifiable models", *The 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*. *2020*

C10. **Raaz Dwivedi**, Nhat Ho, Koulik Khamaru, Martin J. Wainwright, Michael I. Jordan, "Theoretical guarantees for EM under misspecified Gaussian mixture models", *Advances in Neural Information Processing Systems (NeurIPS)*. *2018*

C11. **Raaz Dwivedi**, Yuansi Chen, Martin J. Wainwright, Bin Yu, "Log-concave sampling: Metropolis-Hastings algorithms are fast", Extended abstract in *Conference on Learning Theory (COLT)*. *2018*

C12. Yuansi Chen, **Raaz Dwivedi**, Martin J. Wainwright, Bin Yu, "Vaidya walk: A sampling algorithm based on the volumetric barrier", *Allerton Conference*. *2017*

C13. **Raaz Dwivedi**, Vivek Borkar, "Removing sampling bias in networked stochastic approximation", *International Conference on Signal Processing and Communications (SPCOM)*. *2014*

Journal Publications

J1. **Raaz Dwivedi**, Katherine Tian, Sabina Tomkins, Predrag Klasnja, Susan Murphy, Devavrat Shah, "Counterfactual inference in sequential experimental design", *Annals of Statistics (AoS).* *2025*

J2. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey, "Cheap permutation testing", *In major revision at Annals of Statistics (AoS).* *2025*

J3. Nhat Ho, Koulik Khamaru, **Raaz Dwivedi**, Martin J. Wainwright, Michael I. Jordan, Bin Yu, "Instability, computational efficiency, and statistical accuracy", *Journal of Machine Learning Research (JMLR).* *2025*

J4. **Raaz Dwivedi**, Lester Mackey, "Kernel thinning", Extended abstract in *Journal of Machine Learning Research (JMLR).* *2024*

J5. Raphael Kim, Susobhan Ghosh, Prasidh Chhabria, **Raaz Dwivedi**, Peng Liao, Kelly Zhang, Predrag Klasnja, Susan Murphy, "Did we personalize? Assessing personalization by an online reinforcement learning algorithm using resampling", *Machine Learning Journal.* *2024*

J6. **Raaz Dwivedi**, Chandan Singh, Bin Yu, Martin J. Wainwright, "Revisiting minimum description length complexity in overparameterized models", *JMLR.* *2023*

J7. Nick Altieri, Rebecca L. Barter, James Duncan, **Raaz Dwivedi**, Karl Kumbier, Xiao Li, Robert Netzorg, Briton Park, Chandan Singh, Yan Shuo Tan, Tiffany Tang, Yu Wang, Chao Zhang, Bin Yu, "Curating a COVID-19 data repository and forecasting county-level death counts in the United States", *Harvard Data Science Review (HDSR).* *2021*

J8. **Raaz Dwivedi**, Yan Shuo Tan, Briton Park, Mian Wei, Kevin Horgan, David Madigan, Bin Yu, "Stable discovery of interpretable subgroups via calibration in causal studies", *Int. Statistical Review.* *2020*

J9. **Raaz Dwivedi**, Nhat Ho, Koulik Khamaru, Martin J. Wainwright, Michael I. Jordan, Bin Yu, "Singularity, misspecification, and the convergence rate of EM", *Annals of Statistics (AoS).* *2020*

J10. Yuansi Chen, **Raaz Dwivedi**, Martin J. Wainwright, Bin Yu, "Fast mixing of Metropolized Hamiltonian Monte Carlo: Benefits of multi-step gradients", *Journal of Machine Learning Research (JMLR).* *2020*

J11. **Raaz Dwivedi**, Yuansi Chen, Martin J. Wainwright, Bin Yu, "Log-concave sampling: Metropolis-Hastings algorithms are fast", *Journal of Machine Learning Research (JMLR).* *2019*

J12. **Raaz Dwivedi**, Ohad N. Feldheim, Ori Gurel-Gurevich, Aaditya Ramdas. "The power of online thinning in reducing discrepancy", *Probability Theory and Related Fields (PTRF).* *2019*

J13. Yuansi Chen, **Raaz Dwivedi**, Martin J. Wainwright, Bin Yu. "Fast MCMC sampling algorithms on polytopes", *Journal of Machine Learning Research (JMLR).* *2018*

J14. Vivek Borkar, **Raaz Dwivedi**, Neeraja Sahasrabudhe. "Gaussian approximations in high dimensional estimation", *Systems & Control Letters.* *2016*

Softwares & Methodologies

S1. Albert Gong, Kyuseong Choi, Abhineet Agarwal, Jason Schechner, Ryan Huang, Raj Agrawal, Anish Agarwal, **Raaz Dwivedi**. Benchmark "ORCA-bench" (link). *2026*

S2. Annabelle Michael Carrell, Albert Gong, Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey. Python package "Thinformer" (link). *2025*

S3. Annabelle Michael Carrell, Albert Gong, Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey. Code release "KH-SGD" (link). *2025*

S4. Caleb Chin, Aashish Khubchandani, Harshvardhan Maskara, Kyuseong Choi, Jacob Feitelberg, Albert Gong, Manit Paul, Tathagata Sadhukhan, Anish Agarwal, **Raaz Dwivedi**, "N^2: A unified python package and test bench for nearest neighbor-based matrix completion", *CODEML Workshop, ICML.* *2025*

S5. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey. Python package "Compress then test" (link).

S6. Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey. Python package "Compress++" (link).

S7. **Raaz Dwivedi**, Lester Mackey. Python package "Kernel Thinning" (link).

S8. **Raaz Dwivedi**, Yan Shuo Tan, Briton Park, Mian Wei, Kevin Horgan, David Madigan, Bin Yu. Python repository "StaDISC" (link).

S9. Yuansi Chen, **Raaz Dwivedi**, Martin Wainwright, Bin Yu. Python package (with C++ implementation) "Vaidya and John walks" (link).

Selected Invited Talks

AI Agents for Root Cause Analysis

- MIT Professional Education class *Feb 2026*
- Bridgewater AI Group *Oct 2025*
- Causal inference workshop, ACM Sigmetrics *Aug 2025*

Alumni Panel, 70th Anniversary Statistics Department, BSTARS 2025, UC Berkeley *Sep 2025*

Integrating Double Robustness Into Causal Latent Factor Models

- Informs, Seattle *Oct 2024*
- Joint Statistical Meeting (JSM), Toronto *Aug 2024*
- ESIF conference on Economics and AI+ML *July 2024*
- Mini Workshop on Individualized Decisions, Simons/UC Berkeley *July 2024*
- Tom Ten Have Symposium on Mental Health Statistics, Weil Cornell *Jun 2024*
- Online Causal Inference Seminar *May 2024*
- Operations Research Seminar, MIT *Apr 2024*
- Workshop on Statistical Methods for Digital Health, John Hopkins University *Mar 2024*
- Statistics Seminar, Columbia *Mar 2024*
- Rising Stars in AI, KAUST *Feb 2024*

From HeartSteps to HeartBeats: Personalized Decision-making

- Large Scale Learning and Control Workshop, IIT Bombay *Dec 2023*
- AI Seminar, Cornell University *Sep 2023*
- ORIE Industry and Data Science Summit, Cornell University *Sep 2023*
- Statistics and Data Science Seminar, Cornell University *Sep 2023*
- Center for Applied Math Colloquium, Cornell University *Sep 2023*
- Gatsby Unit Seminar, University College London *Feb 2023*
- Statistics and Data Science Seminar, Yale University *Feb 2023*
- Computer Science Seminar, UIUC *Feb 2023*
- Statistics Seminar, UW Madison *Jan 2023*
- Operations, Information, and Technology Seminar, GSB, Stanford University *Jan 2023*
- Statistics and Data Science Seminar, Wharton, University of Pennsylvania *Jan 2023*
- Statistics Seminar, University of Chicago *Jan 2023*
- Statistics and Operation Research Seminar, UNC Chapel Hill *Jan 2023*
- Statistics Seminar, UCLA *Jan 2023*
- Operation Research and Industrial Engineering Seminar, Cornell University *Dec 2022*
- Operation Research and Industrial Engineering Seminar, Cornell Tech *Dec 2022*
- Statistics Seminar, Rutgers University *Nov 2022*
- ISL Colloquium, EE, Stanford University *Nov 2022*
- BLISS Seminar, EECS, UC Berkeley *Nov 2022*

Compress then test: Powerful kernel testing in near-linear time

- Joint Statistical Meeting, Toronto *Jun 2023*
- Monte Carlo Methods Conference, Paris *Jun 2023*
- Computational-Statistical Interplay in Machine Learning Workshop, MIT *May 2023*

Doubly robust nearest neighbors for counterfactual inference

- Causal Inference Workshop, ACM Sigmetrics, Orlando *Jun 2023*
- New England Statistics Symposium, Boston University *Jun 2023*
- Informs Annual Meeting, Indianapolis *Oct 2022*

Counterfactual inference in sequential experiments

- Institute of Mathematical Statistics (IMS) Annual Meeting, London *Jun 2022*
- Learning from Interventions Workshop, Simons Institute, Berkeley *Feb 2022*

Near-optimal compression in near-linear time

- SIAM Conference on Uncertainty Quantification, Atlanta *Apr 2022*
- Statistical learning Workshop, Mathematical Sciences Research Institute, Berkeley *Mar 2022*

Kernel thinning

- Data-Centric Engineering Group, Alan Turing Institute, Virtual *Sep 2021*

Revisiting minimum description length complexity in overparameterized models

- Alg. Info Theory & Machine Learning Symp., Alan Turing Institute, London *Jul 2022*
- Collaborations on the Theoretical Foundations of Deep Learning, Virtual *Nov 2021*

StaDISC: Stable discovery of interpretable subgroups via calibration

- Young Data Scientist Research Seminar, ETH Zurich, Virtual *Sep 2020*
- ASA Annual Symposium on Data Science & Statistics, Virtual *Jun 2020*

Singularity, misspecification, & the convergence rate of EM

- Math & Statistics Seminar, IIT Kanpur *Jan 2020*
- AMS Special Sections Meeting, UC Riverside *Nov 2019*

Theoretical guarantees for MCMC algorithms

- BIDS Seminar, UC Berkeley *Mar 2019*
- EE Seminar, IIT Bombay *Jan 2018*
- STCS Seminar, TIFR Bombay *Jan 2018*

PhD Students

Kyuseong Choi, Sixth Year, Statistics *2024–*

Brian Cho, Fifth Year, ORIE *2024–*

Albert Gong, Third Year, CS *2024–*

Minor Committee Member

Navonil Deb, PhD → Postdoctoral Fellow, National Library of Medicine (NIH) *2026–*

Tathagata Sadhukhan, Fourth Year, Statistics *2024–*

Teaching Experience

T1. Statistical Principles for Data Science Inference (ORIE 5180), *Cornell University* *Fall 2024*

T2. Causal Inference (ORIE 7790), *Cornell University* *Spring 2024*

T3. Instructor: Statistical Principles (ORIE 6700), *Cornell University* *Fall 2023*

T4. Instructor: Statistical RL for real life (one week; link), *CDT Summer School, Missenden* *Jul 2023*

T5. TA: Sequential Decision Making (STAT 234), *Harvard University.* Gave four guest lectures and supervised several half-semester long research projects. *Spring 2022*

T6. TA: Modern Statistical Prediction and Machine Learning (STAT 154), *UC Berkeley*. Gave one guest lecture and helped in redesign of the class. *Spring 2019*

T7. TA: Introduction to Machine Learning (EECS 189), *UC Berkeley*. Co-head for the content developments in team of 10+ TAs, helped design discussion sections, homeworks, and exams. *Spring 2018*

T8. TA: Linear Algebra, Calculus, Differential equations (MA 105, 106, 108, 207), *IIT Bombay*. Taught teaching sections and several voluntary help sessions that were often attended by 200+ students. *2011–2014*

Academic Services

Undergraduate Research Mentoring

- UC Berkeley, One student that led to a co-authored journal publication *2020—2021*
- Harvard, Two students with three co-authored submissions in preparation *2022—*

Institutional Mentoring

- MIT Institute for Data, Systems, & Society (IDSS) Postdoc Mentors for two *PhD* students *2022—*
- UC Berkeley Artificial Intelligence Research (BAIR) Buddies for two *incoming PhD* students *2020—2021*
- UC Berkeley BAIR Mentoring Program for five *undergraduates* *2017—2021*
- IIT Bombay Student Mentoring Program (ISMP) for twelve *incoming undergraduates* *2013—2014*
- IIT Bombay Academic Mentoring Program (DAMP) for four *sophomores & juniors* *2012—2014*
- IIT Bombay Intensive Mentoring Program for thirty *undergraduates* *2012—2013*

Committees

- Member, Committee on Equality and Diversity, IMS *2022—*

Scientific Meetings

- Organizer and chair, Informs Session on Reinforcement learning *2024*
- Organizer and chair, Informs Session on Causal inference *2024*
- Organizer and chair, Informs Session on Causal inference and reinforcement learning *2023*
- Organizer and chair, Informs Session on Statistical Methods for Healthcare *2023*
- Mentor, Let-All Mentoring Session, Learning Theory Mentorship Workshop *2023*
- Moderator, Panel Discussion on Mentoring, New Researcher Conference Statistics, Toronto *2023*
- Chair, New Researchers Group Session, IMS Annual Meeting *2022*
- Chair, Statistical Machine Learning Session, IMS Annual Meeting *2022*
- Mentor, Summer Institute on Just-in-Time Adaptive Interventions via MRTs *2021*

Graduate Admissions

- ORIE Graduate Admissions Committee, Cornell *2024*
- ORIE Graduate Admissions Committee, Cornell *2023*
- EECS Graduate Admissions Committee, MIT *2021*
- EECS Graduate Admissions Committee, UC Berkeley *2018—2020*

Reviewing Activities

- *Journals*:MOR, OR, Sto. Sys., AOS, JMLR, IEEE-IT, JRSSB, Bernoulli, HDSR, Stats-Comp., SIAM, Jour. of Causal Inference, ISR, JCGS, ACM
- *Conferences*: NeurIPS (Area Chair), COLT, ICML, AISTATS, FOCS, STOC, SODA, AAAI, UAI, SIGMETRICS

Work Experience	**Microsoft Research**, Research Intern with Lester Mackey, New England, USA	*2019*
	Mist Systems, Juniper Networks, Data Science Intern, Cupertino, USA	*2017*
	WorldQuant Research, Senior Quantitative Researcher, Mumbai, India	*2014—2015*
	Stanford University, Research Intern with Prof. Balaji Prabhakar, USA	*2013*
	Ivy Mobility, Data Science Intern, Chennai, India	*2012*